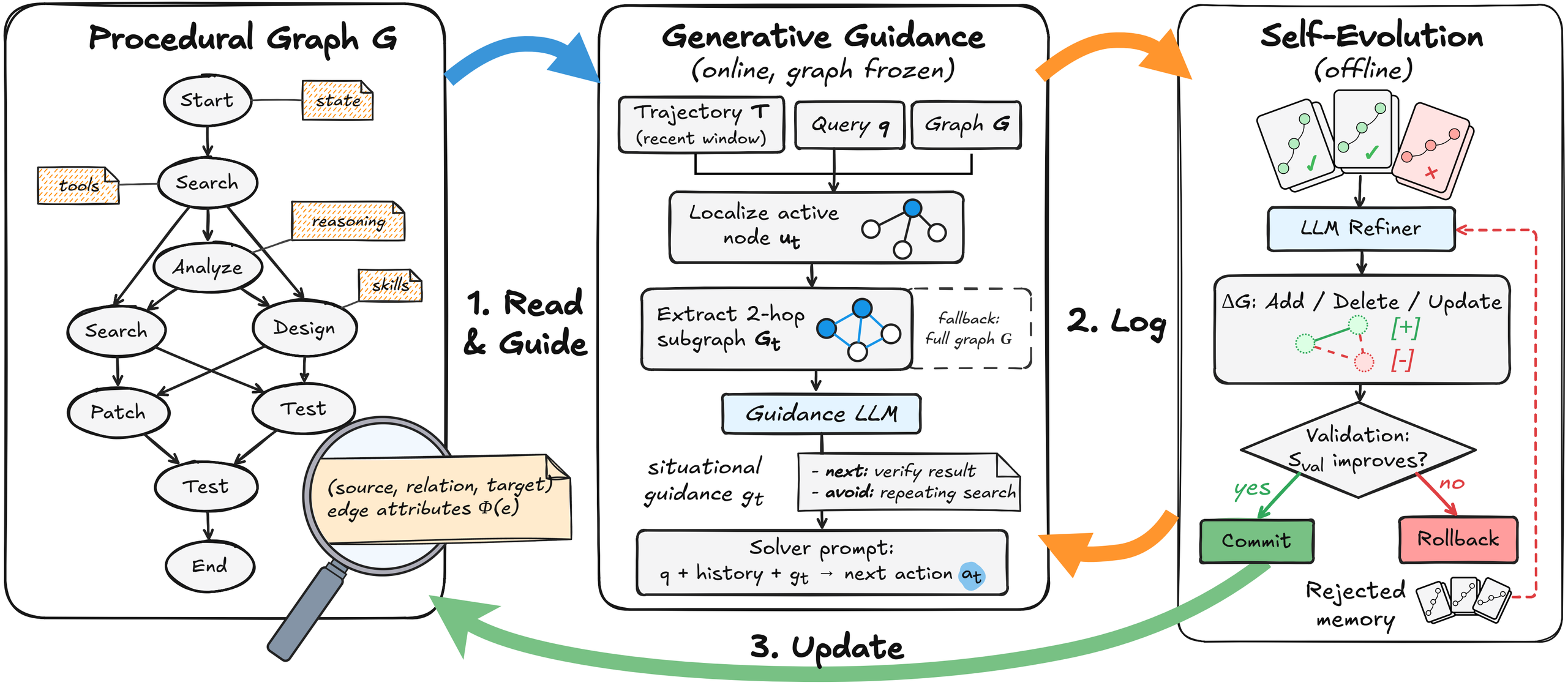

Procedural Graph G
Start
state
tools
Search
reasoning
Analyze
skills
Search
Design
Patch
Test
Test
End
(source, relation, target)
edge attributes Φ(e)
1. Read & Guide
Generative Guidance
(online, graph frozen)
Trajectory T
(recent window)
Query q
Graph G
Localize active node u_t
Extract 2-hop subgraph G_t
fallback: full graph G
Guidance LLM
situational guidance g_t
- next: verify result
- avoid: repeating search
Solver prompt:
q + history + g_t → next action a_t
2. Log
Self-Evolution
(offline)
LLM Refiner
ΔG: Add / Delete / Update
[+]
[-]
Validation:
S_{val} improves?
yes
no
Commit
Rollback
Rejected memory
3. Update